

Tokens Are Multiplied by Turns:

Measuring Context Residency, Reuse, and Reducibility in Coding Agents

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Abstract

Token counts understate the cost of tool-using language-model agents. An input admitted on turn a can remain in the prompt and be processed again on every later turn, so its effective burden is its length multiplied by its remaining lifetime. We call this burden *context residency* and its unit the token–turn. We develop an accounting method that reconstructs real API calls from multi-record transcripts, separates context occupancy from provider-priced cost, attributes resident content to prompt and tool channels, and distinguishes retrospective opportunity from interventions that can be applied prospectively. Three complementary design-partner datasets ground the analysis: an exploratory archive of approximately 168 Claude Code sessions, a preregistered and stratified 50-task Django/SWE-bench-style observation corpus, and the 60 single-window sessions of a live search-reduction study (20 native and 40 reducer runs on four Django tasks), reused for prefix decomposition. In the exploratory archive, whose aggregates are not committed, cache re-reads account for 96% of occupancy. In the 60-session decomposition, the fixed system/tool/injected prefix accounts for 73.0% of resident input token–turns, while retained hidden thinking accounts for an estimated 11.3%. In the frozen 50-task corpus, restricted to the 30 runs that its admission rule makes eligible for token-share claims, 20.9% of fully measured read tokens are retrospectively identifiable exploration candidates and 48.5% form a broader ceiling when search/listing output is included; over all 50 runs the values are 21.3% and 51.0%, with run-bootstrap 95% intervals of 15.9–27.0% and 43.7–58.8%. These are candidate masses of raw read tokens, not token–turns and not guaranteed savings. A sequence of negative results—zero voluntary use of a semantic read surface, graph retrieval that fails a joint recall–budget gate, and eager discovery packets that increase pooled input 55.6%—shows why token compression ratios alone are insufficient. The resulting measurement discipline motivates cache-aware admission and lifetime control, evaluated separately in the companion systems paper.

1 Introduction

Coding agents repeatedly call a language model while reading files, searching a repository, editing code, running tests, and inspecting failures. The prompt on turn t is usually not a fresh message; it contains a fixed system prefix, tool definitions, injected instructions, and much of the prior trajectory. A 2,000-token tool result admitted

early in a 50-call session can therefore create close to 100,000 input token–turns even if it is never read again by the human. Conversely, removing 2,000 tokens from the last request saves almost nothing. The relevant question is not merely *how many tokens can be shortened?* but *which tokens are present, for how many future requests, and under which cache state?*

This distinction is easy to lose because three denominators coexist. First, a context window limits the number of tokens resident on one request. Second, cumulative input workload sums that occupancy over an agent trajectory. Third, an API bill weights uncached input, cache reads, cache writes, and output differently. A transformation can improve the first quantity while worsening the third if it invalidates a cheap cached prefix and creates an expensive rewrite. Under the one-hour cache tier used in our experiments, a write is priced at $2\times$ base input while a hit is $0.1\times$ [5]. Thus even a semantically safe deletion may have negative short-run economic value.

The literature provides strong solutions to adjacent problems. Prompt-compression methods remove lexical units or learn compact representations [16, 17, 32, 12, 41]. Agent-memory systems page, summarize, or actively compact history [31, 26, 37, 46]. Repository retrievers select code using similarity or graphs [45, 27, 9]. Serving systems reuse and schedule KV state [21, 48, 13, 35]. These methods establish that less or better-selected context can preserve or improve task quality. They do not by themselves tell a client-side coding-agent runtime which admitted object is responsible for cumulative workload, whether its deletion is known safe at decision time, or whether the cache-write penalty dominates future read savings.

This paper supplies that missing empirical layer. It makes four contributions:

1. We define a dual ledger for *occupancy* and *economics*, centered on the token–turn and an object-level residency graph.
2. We provide a trace-reconstruction method that merges records belonging to one API request, excludes sidechains from main-trajectory totals, and explicitly separates measured from estimated token classes.
3. We quantify where coding-agent context accumulates using complementary observational and controlled corpora, including a preregistered 50-task sample and task-cluster uncertainty.
4. We report the failed mechanisms, not only the surviving story. Those results reveal three recurring errors: confusing retrospective labels with prospective policies,

optimizing bytes instead of token-turns, and ignoring cache rewrite economics.

All code, protocols, aggregate artifacts (except D1’s, which were never committed), gateway logs used by the companion paper, and scripts that regenerate the figures and tables are in the repository [7]. The work is a single-provider, single-primary-repository design-partner study. Its measurements are evidence for mechanism and experimental design, not a market-wide prevalence estimate.

2 Accounting Model

2.1 From token counts to residency

Let P_t denote the model-visible input on API call t :

$$P_t = U_t + R_t + W_t, \quad (1)$$

where U_t is uncached input, R_t is cache-read input, and W_t is cache-creation input. The cumulative input workload of a trajectory with T calls is

$$\mathcal{W} = \sum_{t=1}^T P_t. \quad (2)$$

The unit of $\mathcal{W}$ is the token-turn. It is an implementation-facing measure of context work, not a claim about the internal floating-point operations of a proprietary model.

For an object i with size s_i , admitted at call a_i and last resident at call d_i , its idealized residency burden is

$$\mathcal{R}_i = s_i(d_i - a_i + 1). \quad (3)$$

An object may be a tool schema, system-prompt segment, file read, shell output, assistant thought, or edit payload. A context-residency graph records object lineage, re-delivery, supersession, and the calls over which the object remains resident. The graph makes explicit why admission and retirement are dual operations: admission controls s_i and a_i ; retirement controls d_i .

2.2 The economic ledger

Provider billing is a weighted ledger rather than a token total. For price profile $\pi = (\alpha, \rho, \omega, \beta)$, we compute

$$\mathcal{B}_\pi = \sum_t (\alpha U_t + \rho R_t + \omega W_t + \beta O_t), \quad (4)$$

where O_t is model output. The experiments in the companion paper use (1, 0.1, 2, 5), matching the one-hour cache-write, cache-hit, and output ratios of the selected model at collection time. We retain the vector (U, R, W, O) in every artifact so that results can be repriced rather than frozen to one vendor schedule.

The two ledgers answer different questions. $\mathcal{W}$ approximates context-window pressure and input processing across the trajectory. $\mathcal{B}_\pi$ answers what the observed usage would cost under π . A policy is not declared economically beneficial merely because it reduces $\mathcal{W}$.

2.3 Prospective and retrospective knowledge

At analysis time, future events are visible. At runtime, they are not. We therefore distinguish:

- **Prospective controls** use only information available before sending the current request: content type, age, cache state, known supersession, and configured tool permissions.
- **Retrospective ceilings** may use the remainder of the completed trajectory to ask what would have been removable by an oracle.

This distinction is central. A file read that is never followed by an edit to the same path is a retrospective exploration candidate. The observation does not prove that the read was semantically unnecessary for a later edit elsewhere, nor could a runtime have known its future role when the read first arrived. We use the word *ceiling* whenever either limitation applies.

3 Relation to Prior Research

The most useful comparison is by intervention boundary rather than by headline compression ratio. Table 1 summarizes the boundary and the remaining gap.

3.1 Prompt compression and long-context behavior

LLMLingua reports up to 20× compression with little loss on several prompting scenarios; LongLLMLingua targets retrieval ordering and long-context position bias; LLMLingua-2 reframes task-agnostic compression as token classification [16, 17, 32]. RECOMP trains extractive and abstractive compressors for retrieved documents, including selective empty augmentation [41]. Gist tokens and ICAE learn compact latent prompt representations [30, 12]. Li et al. provide a broader taxonomy of hard and soft prompt compression [23].

These works demonstrate that redundancy can be removed, but compression quality is not the only constraint in an agent loop. The compressed artifact can remain resident for dozens of future calls; a query-dependent rewrite can also break a stable cache prefix. Moreover, long advertised windows do not guarantee uniform use of evidence: “lost in the middle” behavior and RULER both motivate evaluating effective, rather than nominal, context [25, 15]. Our contribution is not a new linguistic compressor. It identifies the objects and time intervals for which a compressor, deferrer, or deleter could have system-level leverage.

3.2 Agent memory and active compaction

MemGPT treats context as virtual memory and exposes tiered movement to the model [31]. Recent coding-agent systems bring compaction inside the trajectory. CAT uses a structured workspace and model-called context maintenance, reporting 57.6% solved on SWE-bench Verified [26]; Focus reports 22.7% token reduction on five SWE-bench Lite instances at equal 3/5 success [37]; AutoCompact learns when to compact from supervised and reinforcement-learning signals [46]. StateComp trains a router that decides when history can be compressed, reporting 52.27% fewer total tokens on WorkBuddyBench with task performance preserved, without modeling pricing or caching [38]. A harness-design study spans 176 matched settings, crossing five context-management strategies with four win-

Table 1: Research landscape and the distinct question addressed here. “Cache-aware” means the decision objective explicitly includes prefix validity or provider-tier economics, not merely that the implementation can benefit from a cache.

Family	Representative work	Primary unit optimized	Cache-aware	Boundary relative to this work
Prompt compression	LLMLingua, LongLLMLingua-2, RECOMP, ICAE [16, 17, 32, 41, 12]	tokens in one supplied prompt or retrieved document	usually no	Strong evidence that prompts can be compressed; limited accounting for repeated admission over an evolving agent trajectory.
Agent memory / compaction	MemGPT, CAT, Focus, AutoCompact, State-Comp, harness-design study [31, 26, 37, 46, 38, 10]	remembered semantics, bounded trajectory state, total tokens	partial	Closest semantic relatives. Our focus is measured object lifetime, fail-open mutation, and cache-priced break-even behavior.
Repository retrieval	RepoCoder, GraphCoder, LocAgent [45, 27, 9]	relevance, localization, completion or issue-resolution accuracy	no	Better retrieval can reduce context, but relevance/accuracy does not imply a positive whole-session token or dollar return.
Tool discovery	Re-Invoke, ToolRerank, MCP-Zero [8, 49, 11]	tool recall/ranking and selected-schema count	rarely	Directly supports deferred admission. We measure schema residency in a live coding client and expose a proxy-specific deferral failure.
KV-cache serving	vLLM, SGLang, Prompt Cache, ChunkAttention, Preble [21, 48, 13, 44, 35]	server memory, throughput, TTFT, prefix reuse	yes	Server-side systems decide placement and reuse; this work observes a black-box API and decides whether client-side prompt mutation pays.
Agent prompt caching	Don’t Break the Cache, CAPC, keepalive economics, TokenPilot [28, 34, 20, 40]	provider-priced reuse over multi-turn workloads	yes	Concurrent and highly complementary; TokenPilot already prices gateway-level lifecycle eviction. We add content-object attribution, coding-agent mutation safety, and an intervention lineage from failed mechanisms to live results, with gates pre-registered where noted.

dow budgets on SWE-bench Verified and Terminal-Bench, and finds rule-based elision followed by summarization the most efficient; its costs use OpenRouter rates without prompt-cache pricing [10]. TRACE shows that recurrent compression increases execution instability [29], and CompactionRL trains compaction into open models [22]. The broader memory literature spans storage, retrieval, reflection, and update operations [47]. These works establish rule-based elision and learned timing; what they leave open is the cache-priced denominator that this paper measures.

Our results reinforce this direction but also show why the controller matters. Voluntary adoption can be zero, eager replacement packets can increase whole-session workload, and unaligned deletion can turn cheap reads into expensive writes. We therefore treat semantic policy, object safety, and cache economics as separable gates.

3.3 Code retrieval and graph context

SWE-bench established repository-scale issue resolution as a long-context, environment-interaction problem [18]. SWE-agent showed that the agent-computer interface materially changes performance [42], while Agentless demon-

strated the strength and low cost of a simple localization–repair–validation pipeline [39]. RepoCoder iterates similarity retrieval and generation; GraphCoder and LocAgent use structural graphs for completion or localization [45, 27, 9]. Decision-aware memory cards rank evidence by its expected effect on the agent’s next decision and improve retrieval ranking on SWE-bench Verified [14]; like the graph results below, such gains are measured in retrieval metrics, not in whole-session tokens.

Those results motivate our own graph experiments but do not guarantee a token win. A graph may find useful code yet require an anchor the issue text does not provide, or may spend more tokens to achieve the necessary support recall. We report both failures below. The conclusion is scoped: graphs are valuable representations and can improve localization; they did not clear our joint recall–budget gate as the primary saving mechanism in this Django sample.

3.4 Caching and tool admission

PagedAttention reduces KV-memory fragmentation, SGLang’s RadixAttention reuses shared prefixes, and

Prompt Cache makes reusable prompt modules explicit [21, 48, 13]. ChunkAttention shares prefix chunks; RAG-Cache and CacheBlend reuse knowledge state; Preble schedules shared prompts across devices; Parrot exposes application-level dataflow [44, 19, 43, 35, 24]. These optimize an inference service that can see or control KV state. A client of a proprietary API instead observes billed cache reads and writes and can only shape request content and timing.

Tool retrieval attacks a particularly expensive fixed prefix. Re-Invoke and ToolRerank improve tool retrieval; MCP-Zero constructs a task-specific toolchain [8, 49, 11]; Tool Attention gates tools dynamically and loads schemas lazily on a simulated 120-tool benchmark [33]. Provider tool search similarly defers schemas, keeps deferred definitions out of the cached prefix, and explicitly warns that definitions and accumulated results consume context [6, 1, 4]. Our measurement places this idea inside an end-to-end cost ledger and finds that a custom base URL can disable the client’s normal deferral behavior [2]—an interaction large enough to invalidate an otherwise well-designed experiment.

The distinction between token and dollar savings is recognized elsewhere too. VERA reports 31.5–63.1% fewer cumulative non-cache tokens and notes that this is not a monetary cost [36], and TokenPilot prices gateway-level lifecycle eviction against cache hits and misses [40]. Our dual ledger keeps the two quantities separate for an unmodified coding client at the object level.

4 Data and Measurement Design

4.1 Three complementary datasets

No single dataset supports every claim. We therefore use three layers and preserve their evidence grades.

D1: exploratory design-partner archive. Approximately 168 personal Claude Code sessions and roughly 170,000 logged requests were profiled to locate large mechanisms. This convenience sample includes varied interactive work and is used only for structural observations such as cache-read dominance, rebuild causes, aged-result occupancy, and concentration. The private raw archive is not redistributed, and no aggregate artifact for D1 is committed either: the D1 values reported below are unarchived figures from the original profiling and cannot be re-derived from the repository. We do not use D1 for population inference.

D2: frozen observation corpus. Fifty Django tasks were selected from the 231 Django instances in SWE-bench Verified using a SHA-256 rank fixed before collection. The design balances five gold-patch shape strata—one-line, small, medium, large, and multi-file—at ten tasks each. Selection and execution order use separate namespaces. Each run starts from a pinned base commit in an isolated worktree; failures remain in place. The task run, not an individual read, is the sampling unit. Forty-seven of 50 runs satisfy every canonical admission check. All 50 are retained, and 49 contain substantive reads and edits.

Official grading resolves 40/50 patches.

D3: prefix decomposition set. D3 reuses the 60 single-window sessions of a live search-reduction study on four Django tasks: 20 native runs and 40 runs under a search-output reducer (two configurations of 20). The sessions average 49.5 real API calls and 42,451 startup tokens and support a channel-level decomposition of cumulative input; shares are means of per-session shares. The system prompt is not directly visible in transcripts; its mass is inferred as the residual needed to reconcile the first-call provider usage after observed tool and injected segments are counted. Consequently the fixed-prefix share has a documented hard lower bound of 66.9%, while the best estimate is 73.0%. Invisible retained thinking is estimated as output tokens minus visible output, with no correction factor, on these headless Sonnet runs; it assumes that prior-turn thinking stays resident and is not a measurement on reasoning-heavy sessions.

The companion systems paper adds B6–B8 live and replay datasets. Keeping those results separate prevents intervention outcomes from redefining the measurement taxonomy that selected them.

4.2 Trace reconstruction

Claude Code can record one API response on multiple transcript lines that share a `requestId`; each line can repeat the same usage block. Treating lines as calls inflated both the turn axis and summed usage by about 1.9× in an early harness. The corrected parser groups records by `requestId`, merges their content, and counts the usage once. Main-trajectory analysis follows parent lineage and excludes sidechains unless a study explicitly declares them in scope. The parser then exactly reproduces the CLI-reported usage on validation sessions.

This correction is more than bookkeeping. Residency grows with the number of future calls, so an inflated call index biases the apparent benefit of late retirement. We keep the original corrected notes and commit history to make this failure auditable.

4.3 Read-role evidence

The observation runtime records pre- and post-tool events, mutations, virtual working directory, and causal read-to-edit links. A run is canonical only when calls reconcile, capture errors and pending tools are zero, schemas match, provenance is complete, and no sequential fallback was required. Across D2, 538 of 589 classified reads (91.3%) have fully attributed text-token weights. The remaining 51 are composite or multipath ambiguities; no multimodal read enters the token headline. Every edit-precondition link uses actual event linkage rather than temporal proximity, and 76.6% are at causal distance one.

Roles are intentionally conservative: 18.8% of read events are edit preconditions, 12.2% exploration, 1.7% verification, and 67.2% unknown. Much of “unknown” is search output that the runtime refuses to promote to a file-specific role. Bash coverage is certified clean in only 3 of the 50 runs, all trivially small, and the `config_required` category is unexercised. Those scope gaps remain exclusions rather

than being relabeled to improve the opportunity estimate.

4.4 Estimands and uncertainty

For D2 we report the token-weighted micro ratio, task-level macro mean and median, equal-stratum mean, and per-stratum values. The corpus’s admission rule gates token-share headlines to the 30 runs marked token-share eligible, so the eligible-run ratios are the primary headline and the all-50 ratios are secondary. The micro estimand answers where measured tokens occur in this balanced design; it is not an estimate of natural production traffic. The ratios are shares of raw read tokens, not of token-turns. We resample task runs, not read events, 100,000 times to obtain percentile bootstrap intervals for the two all-50 opportunity ratios. The bootstrap quantifies run-to-run variation inside this corpus; it does not repair single-repository or single-client external-validity limits.

5 Results

5.1 The repeated prefix dominates

The D1 values in this paragraph are unarchived: no committed artifact holds them. In D1, cache re-reads form 96% of context occupancy. Eighty-two percent of cache-write tokens are prefix rebuilds rather than newly appended content; approximately 64% of the rebuild mass is attributed to unavoidable idle-TTL expiry and 24% to addressable client behavior. Tool results that remain resident for more than 50 turns account for about 23% of occupancy. Exact hash-identical re-delivery is only 2.4%, model-switch churn 2.7%, and rewrite amplification 1.3 $\times$. The top 20 sessions contribute 74% of occupancy. These exploratory values establish concentration and mechanism, not cross-user prevalence.

D3 explains why naive cleanup underperforms. Figure 1(a) assigns, on average across sessions, 73.0% of input token-turns to the fixed system/tool/injected prefix. Retained hidden thinking contributes an estimated 11.3%, file-read results 6.8%, Bash results 4.9%, and tool-use inputs 2.7%. Polling turns account for only 0.5% on average. Same-path file refreshes add about 2,103 tokens per session, roughly 1% of cumulative input after their remaining lifetime is considered. Edit `old_string` duplication of a resident read is zero in this sample. Two-thirds of these sessions ran under a search-output reducer, so the read and Bash shares are not purely native.

The economic ledger changes the composition but not the priority. Recomputed from the committed usage records of these sessions (58 of 60 carry one) at one-hour cache prices, cache reads are 97.8% of tokens yet 61.1% of price-weighted cost; cache writes are 1.7% of tokens but 20.9% of cost; output is 0.6% of tokens and 18.0% of cost. A prefix rewrite can therefore dominate the savings from removing a modest result. The strongest lever must reduce a large early object while preserving or deliberately rebuilding the cache.

Finding 1 (observed). Coding-agent context is predominantly a lifetime problem. The fixed prefix is multiplied across calls, and provider cache writes make poorly timed mutation economically discontinuous.

5.2 Read-level opportunity is substantial but not directly actionable

Under the corpus’s admission rule, token-share headlines use the 30 token-share-eligible runs, which hold 100,547 fully measured read tokens: there the conservative ratio is 20.9% and the broad ratio 48.5%. Across all 50 runs, D2 contains 246,686 fully measured read tokens. The conservative bucket contains 52,535 tokens (21.3%): file reads retrospectively labeled as exploration because the same path is not later mutated. The broader ceiling adds 73,360 search/listing tokens, reaching 125,895 tokens (51.0%). The remaining mass comprises 70,726 edit-precondition tokens (28.7%), 44,610 unresolved tokens (18.1%), and 5,455 verification tokens (2.2%). All of these are shares of raw read tokens, not of token-turns; for scale, file-read results make up only 6.8% of resident token-turns in D3.

Over all 50 runs, the run-bootstrap 95% interval is 15.9–27.0% for the conservative ratio and 43.7–58.8% for the broad ratio. Reweighting supports the structural conclusion: conservative micro/macro-mean/macro-median/equal-stratum estimates are 21.3/17.3/9.6/20.4%; broad estimates are 51.0/47.4/44.7/51.3%. The conservative opportunity is right-skewed—22 of 50 tasks (44%) contain no confident exploration candidates and only 25/50 exceed 10%. The broad opportunity is more stable: 47/50 tasks exceed 10%, and every stratum lies between 38.9% and 61.6%.

These ratios are deliberately called ceilings. Search/listing content is identifiable prospectively by type, but a safe summary still needs to preserve filenames, matches, and evidence used later. Exploration is known only retrospectively and same-path non-mutation does not imply semantic irrelevance. The gap between candidate mass and realized savings must be closed by an intervention with task-quality checks.

Finding 2 (observed ceiling). About half of measured raw read tokens fall into a broad reduction-candidate set (48.5% on eligible runs, 51.0% on all 50). Over all 50 runs, only 29.7 percentage points are type-identifiable prospectively; the remaining 21.3 points depend on future knowledge and cannot be advertised as safe live savings.

5.3 Four mechanisms fail for four different reasons

Table 2 collects the program’s negative and boundary results. They are important because each would look promising under a narrower denominator.

Search compression optimizes a small slice. A transparent reducer preserved evidence and removed 12.1% of search-output tokens in a paired replay whose output is not committed. That figure is a search-only ratio and is not the same quantity as an earlier cap model’s 12.1%, which was a share of all read tokens; on the like-for-like search-only basis the cap model predicted 40.8%, about

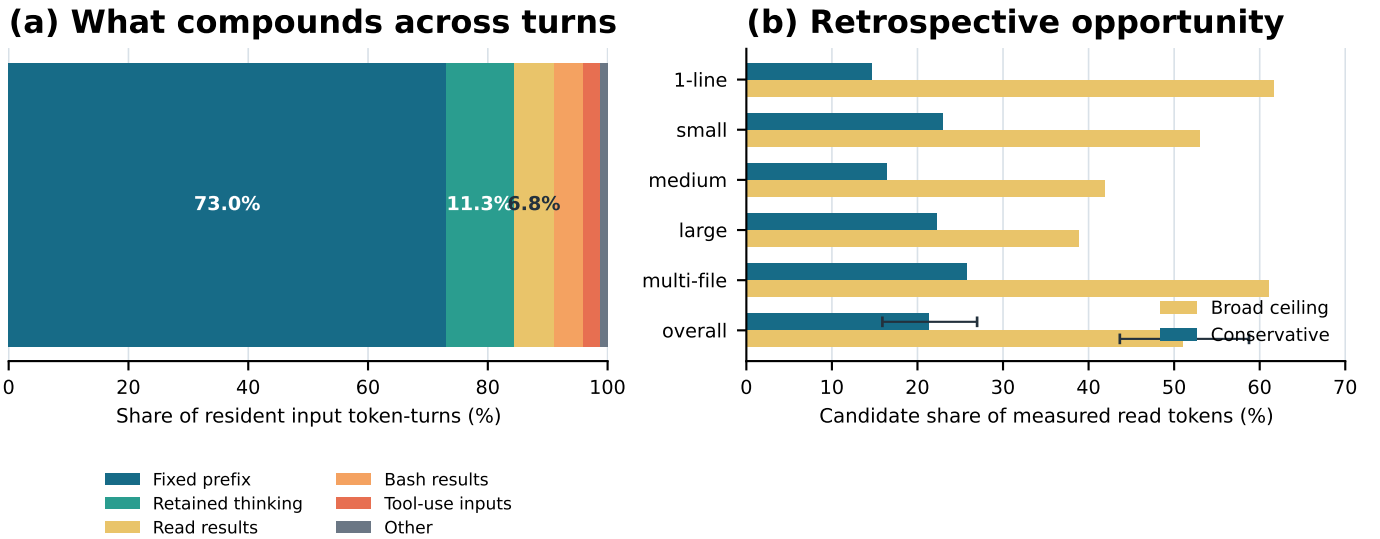


Figure 1: Measured context structure. (a) Channel decomposition across the 60 single-window Django sessions of the live search-reduction study (20 native, 40 under a search-output reducer), as means of per-session shares; retained thinking is estimated and the fixed-prefix estimate has a 66.9% hard lower bound. (b) Candidate raw read-token mass across five balanced fix-shape strata and overall in the frozen 50-run corpus, using all 50 runs; restricted to the 30 token-share-eligible runs the overall values are 20.9% and 48.5%. Overall whiskers are task-run bootstrap 95% intervals. The broad ceiling adds search/listing output to retrospective exploration candidates; neither bar is a guaranteed saving.

3.4× the replay. The replay may itself be biased low, because the replayed transcripts may already contain reduced outputs; we have not verified this. In live replicated runs (three arms of 20 runs on four tasks), the directly removed mass was only 935–1,807 tokens per session, 0.028–0.048% of cumulative tokens. Across 57 completed trajectories, total tokens correlated with the number of turns at $r = 0.988$ ($r^2 = 0.976$). The treatment arms made more turns (pooled +4.9% and +15.8%), so whole-session totals rose (pooled +6.4% and +20.8%) despite genuine per-result compression. Exact repeated searches stayed at zero, but repeated-scope searches rose from 55 in the native arm to 95 and 110 in the two reducer arms, and task success in these runs was not graded. The mechanism preserved evidence locally but was not material.

A correct interface can have zero adoption. A SemanticFS server indexed 43,241 Django symbols, resolved natural names, and returned budget-valid bundles. Two independently written system prompts asked the agent to use the surface. Delivery was logged and forced calls worked, yet voluntary adoption was 0/11. Repository issues often provided concrete paths or tracebacks that competed with a new abstract interface. We closed advisory-only adoption rather than counting potential bundle savings as realized.

Graph recall and token budget form a joint gate. In G1, a problem-statement anchor reached none of 14 edited symbols. Supplying the edited symbol as an oracle anchor improved symbol recall to 0.857 and edit-line/body recall to 0.571 at a 2,048-token budget; the bundle was 41% of native exploration tokens. The result is compression conditional on already knowing the target. In G2, a task-relevance graph had a budget-free support ceiling of 0.739, but at 1,024 tokens its 0.137 recall gain missed

the 0.15 gate; at 2,048 and 4,096 tokens the gain cleared 0.15 only by spending 49% and 195% more tokens than lexical retrieval. No budget cleared both gates. This does not contradict LocAgent or GraphCoder, whose target is localization/completion quality rather than our joint whole-session saving criterion.

An offline call oracle can reverse live. Discovery accounts for 50.4% of calls in a 60-session analysis. An evidence-gated oracle suggested 13.1% of all calls could collapse while retaining 97.1% of next-action evidence, 100% of edit-target evidence, and 96.3% of edit-region evidence. A live executor then enforced the replacement. Agents used it as a search engine, making 15–16 iterative calls per session; each eager multi-kilobyte packet stayed resident. Across three tasks, input rose 11.5%, 39.5%, and 163.8% (mean +71.6%, pooled +55.6%). The enforcement run converted cheap probes into expensive resident packets. We closed the eager-packet design rather than add a learned retriever to rescue the hypothesis.

Finding 3. Compression ratio, retrieval recall, adoption, and cache economics are independent gates. Passing any one does not license an end-to-end efficiency claim.

6 What the Field Has Established—and the Remaining Story

The literature and our evidence now support a cleaner research story than “agents use too many tokens.”

Established: content can be compressed. Textual and latent compressors achieve large ratios on QA, summarization, and long-context tasks [16, 17, 32, 12, 41]. This motivates a compression operator but not where to apply it in a coding trajectory.

Established: managed memory can preserve

Table 2: Mechanism-screening ledger. Gates were written before the corresponding confirmatory run where noted. “Closed” means closed as a primary token-saving lever in this environment, not a claim that the representation or idea is universally useless.

Mechanism	Attractive local metric	Whole-system evidence	Evidence type	Decision / lesson
Search-output compaction	12.1% fewer search-output tokens in a paired replay that may be biased low; the earlier cap model predicted 40.8% on the same basis	Direct saving only 0.028–0.048% of total live tokens; totals track turn count ($r = 0.988$, $r^2 = 0.976$)	Live study: 3 arms $\times$ 20 runs on 4 tasks, 57 completed; success not graded	Evidence-preserving locally, too small in the repeated-prefix denominator; repeated-scope searches rose.
Semantic read surface	Budget-correct symbol bundles; forced calls resolve	0/11 spontaneous uses across two delivered advisory prompts	11 live runs, 6 tasks	Advisory adoption closed; a usable mechanism without adoption has zero product effect.
Graph-first context	At 2,048 tokens, oracle-anchored bundle is 41% of native exploration	Problem text anchors reach 0/14 edited symbols; task-graph budgets never clear recall and token gates together	Offline G1 (14 edits) and G2 (3 tasks)	Graph is a relevance representation, not automatically a token-saving policy.
Discovery call collapse	Offline evidence-gated oracle removes 13.1% of calls at >96% evidence retention	Forced eager packets raise mean input 71.6% and pooled input 55.6%	3 enforced live tasks after offline oracle	Iterative multi-kilobyte packets remain resident; local call reduction does not imply token–turn reduction.

long-horizon capability. Virtual memory, structured workspaces, agent-controlled compaction, rule-based elision, and learned compression timing improve bounded-context operation [31, 26, 37, 46, 10, 38]. This motivates lifetime control but leaves the cache-priced billing objective largely open.

Established: repository structure improves retrieval in some tasks. Iterative and graph retrieval improve completion or localization [45, 27, 9]. Our evidence says the quality objective and a token-saving objective must be evaluated separately, especially when anchors are unknown.

Established: prefix reuse is a systems primitive. Serving systems obtain large throughput or latency gains by reusing KV state [21, 48, 13, 44, 35]. Recent API-level work shows that cache strategy materially changes multi-turn cost [28, 34, 20, 40]. This motivates a cache-aware controller rather than deletion at a fixed token threshold.

The gap: causal, economic object lifetime in real coding agents. The missing layer connects what an object means, how long it will remain, whether it can be changed safely, and what that change does to a black-box provider cache. Our measurement results identify fixed-prefix admission as the largest prospective lever, old tool/thinking state as a tail lever, and cache-aligned timing as the condition that makes deletion economically sane. The companion paper turns that map into a runtime and tests it live.

7 Implications for Evaluation and Design

7.1 Report a claim ladder

Every efficiency number should name its rung:

1. **Candidate mass:** tokens matching a retrospective or type-based rule.
2. **Counterfactual opportunity:** replay under an ex-

plicit behavior/cache model.

3. **Live context reduction:** observed cumulative input under an intervention.
4. **Live economic reduction:** observed or transparently repriced billing components.
5. **Quality-preserving reduction:** the above with objective task evaluation and a frozen acceptance rule.

Moving upward requires new evidence; ratios must not be relabeled. In particular, B7’s 61.5% pooled giant-session result (49.7% over all 54 replayed sessions) in the companion paper remains modeled: its saving comes from 8 interactive sessions, and its cache model is validated only on the append-only branch (exact on 11 of 12 live native sessions, with the twelfth off by +103.7%), never on the edit branch or on a live break-even fire.

7.2 Treat negative results as mechanism localization

The failed sequence was not wasted effort. Zero adoption identifies a policy-interface problem. Graph failure separates retrieval relevance from economic value. Search compression shows that an evidence-preserving local reducer can be immaterial in the global denominator. Eager packets expose residence time as the hidden multiplier. Each result removes a branch from the causal story and motivates the final architecture more strongly than a post-hoc collection of favorable ablations.

7.3 Optimize admission before sophisticated memory

In D3 the fixed prefix contributes 73.0% of observed input token–turns, and tool definitions can be deferred or disallowed before the first request. Old results matter primarily in long-tail sessions. Thus a robust deployment should begin with a prefix audit, remove servers and schemas that cannot be used, retain a small always-on core, and then add

lifetime policies. Provider guidance now recommends the same layered decomposition—tool search for definitions, programmatic calls for round trips, prompt caching for stable content, and context editing for old results [4, 6, 3]. Deferral and admission are thus first-party mechanisms; our contribution is to make the ordering empirical and auditable.

8 Threats to Validity

External validity. The controlled corpora use one repository (Django), one model family (Sonnet), and one coding client (Claude Code); D3 covers only four tasks, and two-thirds of its sessions ran under a search-output reducer. Django is large and real, and SWE-bench tasks require repository navigation, but languages, agents, tools, models, and provider caches differ. D1 is a single-user convenience sample whose aggregates are unarchived. We label all results design-partner evidence and do not extrapolate prevalence.

Construct validity. $\mathcal{W}$ measures billed/model-visible input, not energy, latency, or exact attention FLOPs. Hidden thinking is estimated. The fixed-prefix residual is bounded but not fully observable. The economic ledger uses a stated price profile and should be repriced when provider terms change.

Causal validity. Retrospective same-path labels do not prove semantic dispensability. Replays hold the recorded trajectory fixed and cannot capture behavioral adaptation unless followed by a live experiment. That is why candidate and modeled quantities remain explicitly below live claims.

Sampling and uncertainty. The 50-task corpus is balanced by fix shape rather than sampled in natural traffic proportions. Bootstrap intervals contain only 50 task units. G1/G2 and the eager executor use small diagnostic samples and support scoped mechanism decisions, not universal nulls.

Instrumentation. Raw private design-partner transcripts cannot be released, and the raw transcripts behind the Step-5 to Step-7 studies (including D3) are not in the repository either, so D3 can be checked only against its committed summary and the paired search replay, which has no committed output, not at all. No D1 aggregate is committed. The frozen controlled artifacts, protocols, hashes, request-merging parser, and derived summaries are released. An early request-counting error demonstrates that instrumentation bugs are plausible; preserving the correction and reconciling to CLI totals reduces, but cannot eliminate, this risk.

9 Conclusion

The effective cost of coding-agent context is governed by length, lifetime, and cache state. In our design-partner measurements, repeated fixed content dominates cumulative input, and read-level candidate mass substantially overstates immediately realizable savings. Negative experiments show why: agents may ignore a new interface, graphs may miss a joint recall-budget gate, a locally small reducer may disappear in the whole-session denominator,

and eager replacement packets can remain resident long enough to reverse the expected benefit. A defensible efficiency claim must therefore connect object semantics to future calls and cache economics, then survive a live quality-gated test. The companion paper evaluates that systems path.

Artifact and Reproducibility Note

Code and artifacts: <https://github.com/ankit961/LLM-Token-efficiency>

The repository contains the frozen task plan, observation protocols, aggregate artifacts (D1 excepted), mechanism findings, analysis code, and the exact script that regenerates this paper’s figure and bootstrap intervals; raw session transcripts are not included. Run `python3 papers/scripts/generate_assets.py` from the repository root, then build this source with `latexmk -pdf main.tex`. No paid model calls are needed to reproduce the reported analyses.

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